

## EXPERIMENT 1

AIM: To demonstrate confusion matrix using python

ALGORITHM:

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay, accuracy_score
```

```
import matplotlib.pyplot as plt
```

```
# Actual values
```

```
y_true = [1, 0, 1, 1, 0, 1, 0, 0]
```

```
# Predicted values
```

```
y_pred = [1, 0, 1, 0, 0, 1, 1, 0]
```

```
# Confusion Matrix
```

```
cm = confusion_matrix(y_true, y_pred)
```

```
print("Confusion Matrix:")
```

```
print(cm)
```

```
# Accuracy
```

```
accuracy = accuracy_score(y_true, y_pred)
```

```
print("Accuracy:", accuracy)
```

```
# Display Confusion Matrix
```

```
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=["No Purchase",  
"Purchase"])
```

```
disp.plot(cmap="Blues")
```

```
plt.title("Confusion Matrix")
```

```
plt.show()
```

```

# Confusion Matrix in Google Colab

from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay, accuracy_score
import matplotlib.pyplot as plt

# Actual values
y_true = [1, 0, 1, 1, 0, 1, 0, 0]

# Predicted values
y_pred = [1, 0, 1, 0, 0, 1, 1, 0]

# Confusion Matrix
cm = confusion_matrix(y_true, y_pred)

print("Confusion Matrix:")
print(cm)

# Accuracy
accuracy = accuracy_score(y_true, y_pred)
print("Accuracy:", accuracy)

# Display Confusion Matrix
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=["No Purchase", "Purchase"])
disp.plot(cmap="Blues")

plt.title("Confusion Matrix")
plt.show()

```

## OUTPUT

